

# Approach Document

## 1. Architecture

The system is a conversational retrieval agent with a strict separation between retrieval (deterministic) and generation (LLM-based).

[User POST /chat] —> [State Extractor] —> [Intent Detection] —> [BM25 Search] —>

## 2. Retrieval Design

**Why hybrid retrieval maximises Recall@10:**

Stage 1 — BM25: Okapi BM25 with  $k_1 = 1.5$ ,  $b = 0.75$  over a composite document field (name + description + competencies + technical\_domains + use\_cases). Catches exact keyword signals like "Java", "Docker", "HIPAA", "DSI".

Stage 2 — Metadata Scoring: Five orthogonal signals combined with learned weights:

| Signal               | Weight | Rationale                |
|----------------------|--------|--------------------------|
| BM25 (normalised)    | 0.35   | Keyword recall anchor    |
| Competency overlap   | 0.25   | Role-skill alignment     |
| Test type preference | 0.15   | User expressed needs     |
| Seniority match      | 0.10   | Avoids level mismatch    |
| Use case match       | 0.10   | Selection vs development |
| Language match       | 0.05   | Constraint satisfaction  |

Stage 3 — LLM reranking: Groq Llama 3.3 70b receives the top-10 pre-retrieved candidates and conversation context. LLM selects the optimal 2-8 items, writes rationale, and produces the table.

Stage 4 — Validation: Every URL in the LLM output is checked against the catalog. Any hallucinated item is dropped. Fallback to retrieval results if LLM produces no valid items.

## 3. State Reconstruction

Every API call receives the full conversation history. A deterministic extractor builds a structured state object:

{ "role": "java\_developer", "seniority": "professional", "technical\_skills": ["java", "spring"], "use\_case": "selection", "turn\_count": 4 } This state is the single source of truth. No memory = no state bugs.

## 4. Clarification Policy

**Rules:**

- 1. Never ask more than one clarification round (compound questions only).
- 2. If `turn_count`  $\geq 6$ , force a recommendation to preserve the 8-turn budget.
- 3. Ask *only* when both role and seniority are unknown.

## 5. Evaluation Methodology

Automated test suite (40+ tests targeting 100% pass rates across critical functions):

- **Catalog Integrity & Schema Compliance:** 14 tests ensuring valid payloads and valid SHL URLs.
- **State Extraction:** 10 tests validating accurate constraint parsing.
- **Refusal & Probes:** 11 tests verifying the agent drops illegal/hallucinated prompts.
- **Retrieval Recall:** 10 end-to-end tests measuring Recall@10 against ground-truth scenarios (Target:  $\geq 0.7$ ).

## 6. Tradeoffs & Failed Experiments

**Tried: Pure semantic embedding search** — Lower recall for exact technical terms (Java, Docker). Dropped.

**Tried: Single large prompt with full catalog** — 60 items  $\times$  average 500 tokens = 30k tokens per call, slow and expensive. Switched to pre-retrieval  $\rightarrow$  LLM reranking.

**Tried: Per-question clarification** — Burns turns. Switched to compound single-question policy.

**Tradeoff: No persistent vector store** — For a 60-item catalog, BM25 in-memory is sufficient and instant. At 1000+ items, would add FAISS or pgvector.

**Tradeoff: Groq Llama 3.3 70b** — High-speed API inference, large 128k context window, excellent markdown table instruction compliance.